CDC BRFSS 2015 · RESEARCH PROTOTYPE

Early Diabetes Risk Screening

Can self-reported health answers find adults with diabetes or prediabetes, while keeping missed cases low?

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THE QUESTION

A screening tool should miss few cases without flagging everyone

253,680

adults in the CDC's 2015 BRFSS survey

21

self-reported answers each: health, habits, age, income

15.8%

report diabetes or prediabetes

The bar to clear: not random guessing, but the paper Diabetes Risk Test that clinics already use (NIDDK/ADA).

THE ROADMAP

Eight connected stages, each with its own checks

01
Data ingestion & quality control
Shape, schema, hashes

02
Preprocessing & cleaning
Nothing imputed; odd answers flagged

03
Exploratory data analysis
Balance, distributions, correlations

04
Hypotheses & statistics
25 tests, effect sizes, BH correction

05
Splitting & imbalance
Locked 80/20 split, no weighting

06
Training & tuning
Logistic regression, random forest, XGBoost

07
Evaluation & thresholds
Cutoff locked, test set scored once

08
Reporting & handoff
Report, notebooks, slides, video

Stages 2 to 7 (shaded) are the analysis; every following slide is labelled with its stage.

STAGE 1 · DATA INGESTION & QUALITY CONTROL

253,680 rows × 22 columns, checked; prediabetes counts as positive

Survey status	Respondents	Main analysis	Prediabetes only
No diabetes	213,703	negative	negative
Prediabetes	4,631	positive	positive
Diabetes	35,346	positive	excluded

Checked: shape, column names, data types, value ranges and labels, plus SHA-256 hashes so anyone can confirm the same file.

Main: 39,977 positives (15.8%). **Sensitivity check:** prediabetes alone, 4,631 of 218,334 (2.1%).

Nothing imputed or dropped; odd answers flagged, not changed

0

missing or out-of-codebook values; the pipeline stops rather than guess

23,968

rows repeat another person’s answers; kept, since no ID proves they are copies

14 · 4 · 3

yes/no answers · ordered bands (age, health, education, income) · counts (BMI, health days)

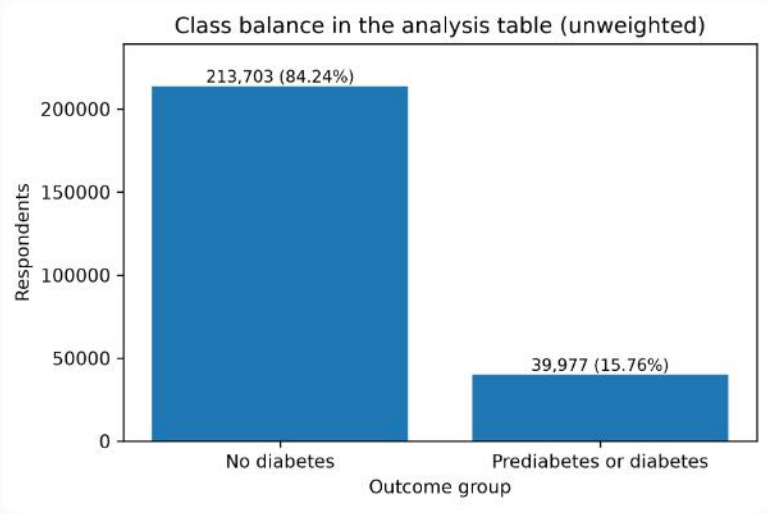
Unusual but valid answers, counted

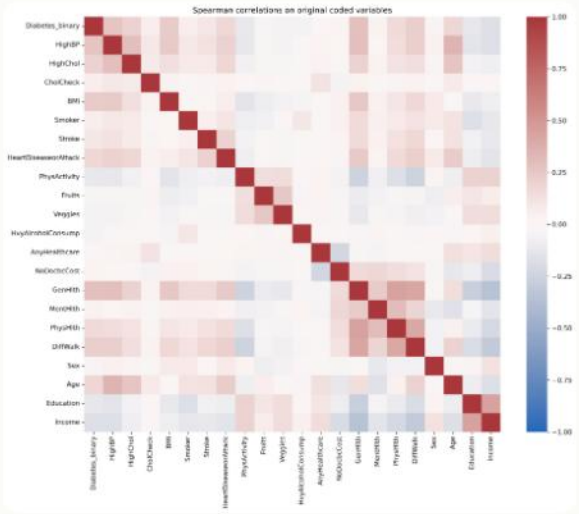
Review flag	People
High cholesterol, no check in 5 years	1,981
BMI below 15	68
BMI 60 or above	868
30 of 30 poor mental-health days	12,088
30 of 30 poor physical-health days	19,400

BMI: median 27 (IQR 24–31). No scaling here: it happens inside the model pipelines (stage 5).

STAGE 3 · EXPLORATORY DATA ANALYSIS

One adult in six is positive; five answers carry the signal





Strongest correlations with the outcome

General health 0.30
High blood pressure 0.27
BMI 0.24
Difficulty walking 0.22
High cholesterol 0.21

Positive group: median BMI 31 vs 27, median age 65–69 vs 55–59.

Accuracy would mislead: answering “no” for everyone scores 84%. Several strong answers move together, so stage 6 adjusts for them.

Spearman correlations on all 253,680 respondents; plots in reports/uploaded-combined/eda · 6

All 25 tests significant, so we read effect sizes

Chi-square: odds ratio, yes vs no (95% CI)

Answer	Odds ratio
High blood pressure	4.78 (4.67–4.90)
Difficulty walking	3.70 (3.61–3.79)
High cholesterol	3.24 (3.17–3.32)
Physically active	0.50 (0.48–0.51)
Cholesterol check*	5.87 (5.23–6.58)

Mann–Whitney U: rank-biserial r

Answer	Effect (95% CI ±0.006)
General health	0.450
BMI	0.372
Age band	0.293
Income band	−0.266
Education band	−0.189

18 chi-square and 7 Mann–Whitney tests, Benjamini–Hochberg corrected. Unadjusted associations, not causes. *A cholesterol check mostly reflects contact with doctors.

STAGE 5 · SPLITTING & IMBALANCE

Every decision used training rows; the test set stayed locked

Training: 202,944 people, five fixed stratified folds for tuning and the cutoff

Test: 50,736



No leakage by design

Robust/standard scaling sits inside the model pipelines, learned from training folds only.

Repeated answers: tested

15% of people share answers. Grouping them changed PR-AUC by 0.001 at most.

Weighting vs none

Same PR-AUC (±0.001), but risks inflated from 15.8% to 39%. No weighting; the cutoff is set in stage 7.

STAGE 6 · TRAINING & TUNING

Every tuned model beats the checklist; XGBoost wins every fold

Validation, mean of 5 folds	PR-AUC	ROC-AUC	Flagged for 80%
No model	0.158	0.500	100%
Paper checklist	0.325	0.756	51%
Logistic regression	0.439	0.819	39%
Random forest	0.456	0.822	39%
XGBoost (chosen)	**0.461**	**0.826**	**38%**

Tuned by randomized search on PR-AUC

Logistic regression: C = 3.47, bands one-hot

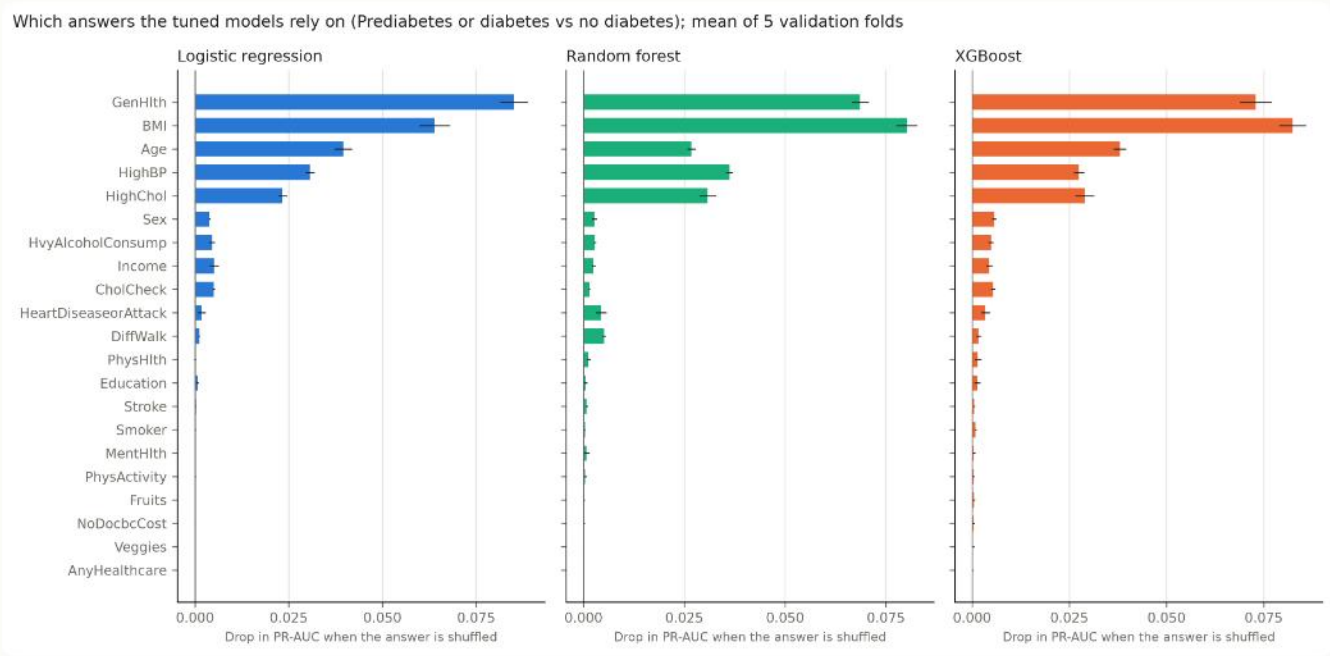
Random forest: 200 trees, depth 12, ≥10 per leaf

XGBoost: 800 trees, depth 3, learning rate 0.058

Rule set before the run: best PR-AUC unless logistic regression is not significantly worse. XGBoost won 5 of 5 folds ($p < 0.001$ vs LR, $p = 0.005$ vs forest).

Stratified 5-fold cross-validation, 12 / 8 / 16 candidates · comparison.csv and best_params.json · 9

The same five answers drive every model



statsmodels: adjusted odds ratios

Physical-health days: 1.045 → 0.993, reversed

Smoking: 1.40 → 0.98, gone

Physically active: 0.50 → 0.95

Once age, BMI and general health are held fixed. All VIFs below 1.8.

STAGE 7 · EVALUATION & THRESHOLDS

We locked the cutoff before opening the test set

0.153 predicted risk; anyone at or above it is flagged

1. Out-of-fold risks on training rows only (reproduce stage 6: 0.4609 vs 0.4610)
2. Highest cutoff that still flags 80% of training cases
3. Calibration rule fixed in advance: slope 0.992, so no recalibration
4. Saved to threshold.json; rerunning gives an identical file

What other targets cost

Catch	Flagged per 1,000
70% of cases	294
80% (locked)	**380**
90% of cases	510

STAGE 7 · FINAL RESULT · 50,736 PEOPLE THE MODEL NEVER SAW

Most cases found, far fewer false alarms than the checklist

79.4%

of cases found (95% CI 78.5–80.3%)

37.6%

of people flagged; 1 in 3 of them has the condition

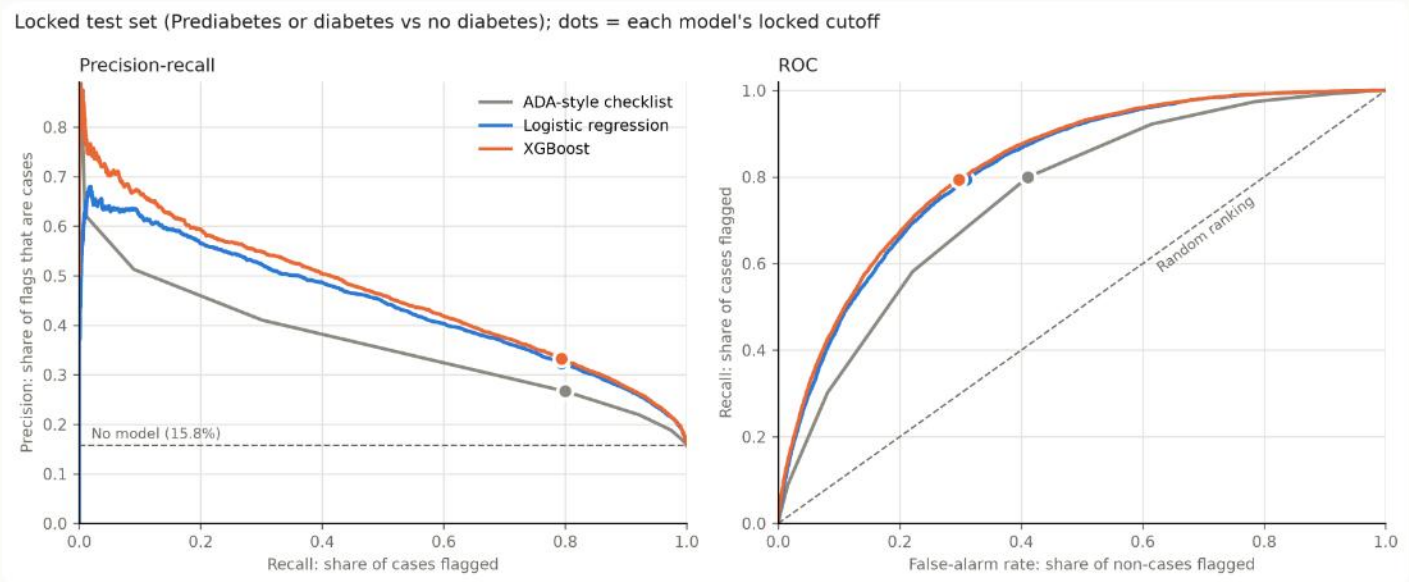
−96

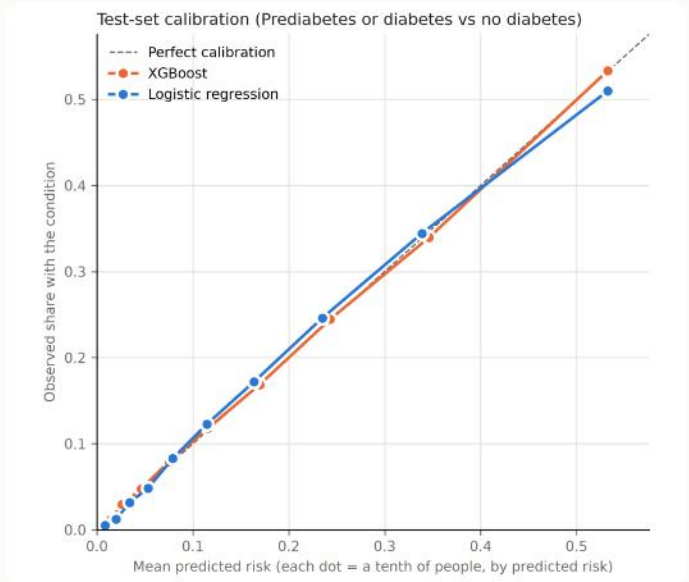
false alarms per 1,000 vs the paper checklist

Precision 0.333 · specificity 0.702 · F2 0.622 · ROC-AUC 0.826 · PR-AUC 0.465, matching validation

STAGE 7 · PR/ROC CURVES & CALIBRATION

Above the checklist at every cutoff, with honest risks

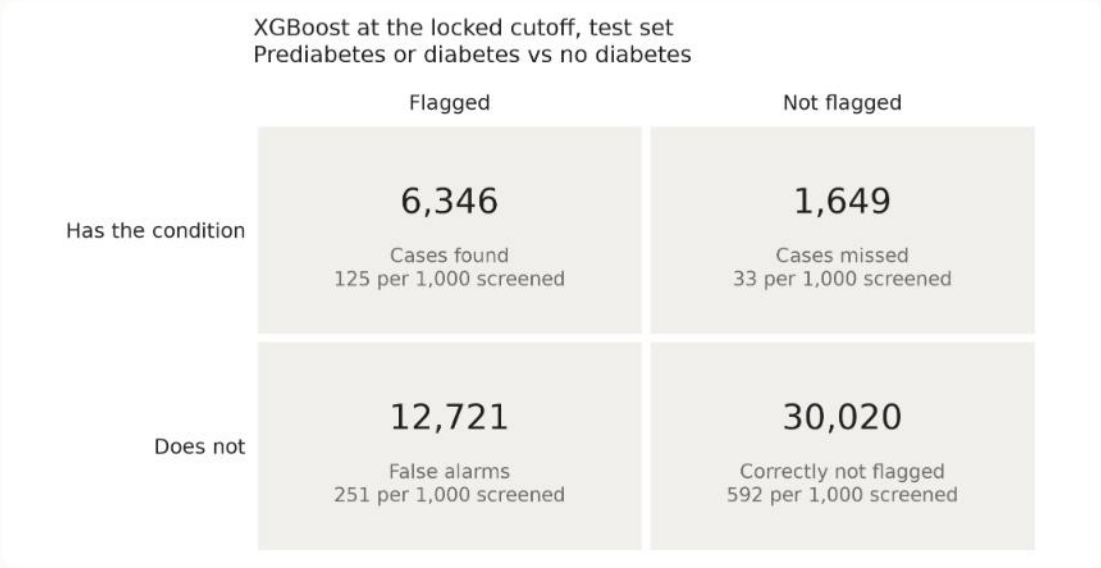




Dots mark each model’s locked cutoff. Right: test-set risks match observed rates (slope 0.991, Brier 0.105).

STAGE 7 · CONFUSION MATRIX & WORKLOAD

Per 1,000 screened: 125 cases found, 33 missed

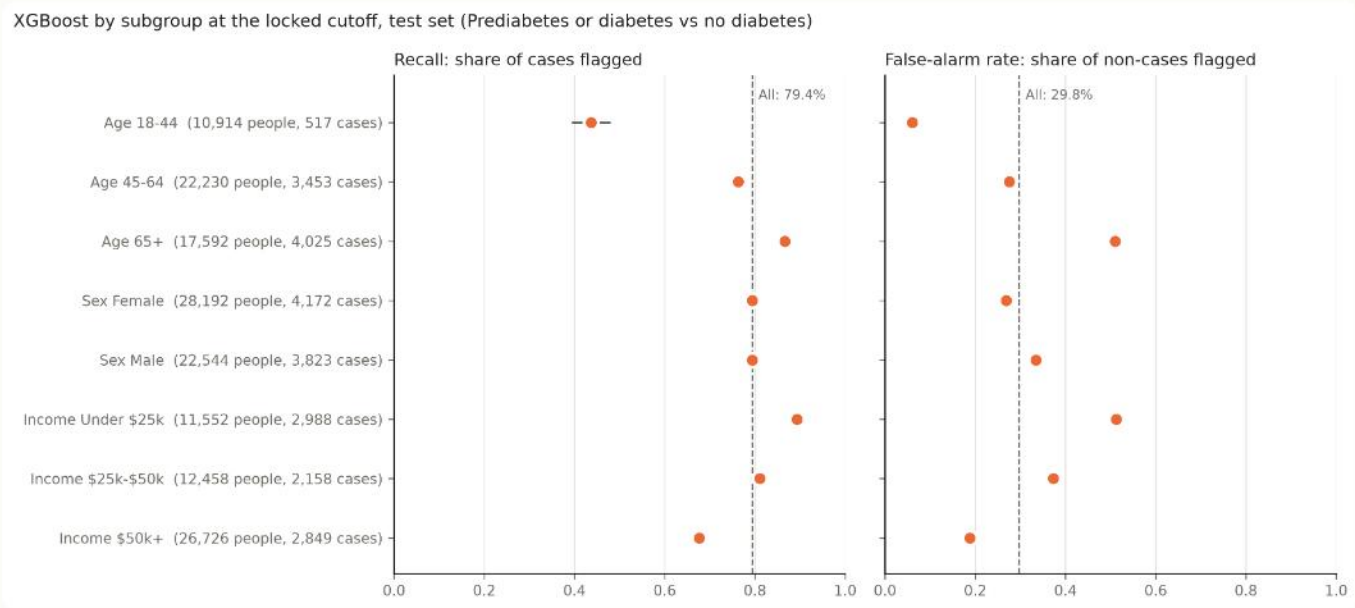


Per 1,000	Flagged	False alarms
Checklist	473	347
Logistic reg.	386	261
XGBoost	**376**	**251**

All three find about 125 of the 158 cases. Specificity: 0.702 for XGBoost against 0.589 for the checklist.

STAGE 7 · FALSE NEGATIVES: WHO GETS MISSED

It misses younger and better-off people, and prediabetes



Diabetes: 81% found

Prediabetes: 65%

The 1,649 missed cases look like non-cases on paper: BMI 27.5 vs 27.7, fair or poor health 9% vs 13%.

STAGE 7 · SENSITIVITY ANALYSIS: PREDIABETES ONLY

For prediabetes alone, the signal is weak

81%

of prediabetes found (checklist: 70%)

44%

of everyone flagged to get there

4%

of flagged people report prediabetes

218,334 people without diagnosed diabetes, 2.1% with prediabetes; PR-AUC 0.062 against 0.021 for no model. Many “false alarms” may be undiagnosed prediabetes, which this survey cannot show.

STAGE 8 · LIMITATIONS

A research prototype, not a diagnostic tool

Self-reported, one moment

Identifies current, known status; predicts nothing about the future.

88% already diagnosed

The model partly learns the footprint of a diagnosis.

Doctor-contact answers

Without them: −0.03 PR-AUC, still well above the checklist.

Unequal trade-offs

One cutoff: 44% sensitivity under 45, 51% false alarms over 65.

Unweighted, one year

No national estimates; untested on other years or lab results.

Associations only

Odds ratios, adjusted or not, are not causes.

STAGE 8 · SOURCES

Data sources and reading

1. **Kaggle, Diabetes Health Indicators** (the data): kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset
2. **UCI, CDC Diabetes Health Indicators**: archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators
3. **CDC, MMWR 2017 (66:43)**: cdc.gov/mmwr/volumes/66/wr/mm6643a2.htm
4. **CDC / NCHS, NHANES**: cdc.gov/nchs/nhanes
5. **CDC / NCHS, Series 11 No. 237** (historical; check its date before citing): cdc.gov/nchs/data/series/sr_11/sr11_237.pdf
6. **NIDDK Diabetes Risk Test** (the checklist baseline): niddk.nih.gov

The Kaggle and UCI files derive from the 2015 BRFSS: self-reported indicators, not a clinical diagnosis workflow.

STAGE 8 · TAKEAWAY

Survey answers can find 8 in 10 cases with a fifth fewer people sent for testing than today's paper checklist

Reproducible

One command per stage; the prepared data, split, cutoff and test tables rebuild identically. 63 automated tests.

In the repository

Code, seven stage notebooks, every report and the technical PDF: github.com/jadonsinghsumit/early-diabetes-risk-screening

Thank you. Questions?